

Quantifying and Attributing Polarization to Annotator Groups

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Abstract

Current annotation agreement metrics are sensitive to group size imbalances and noise, rendering them insufficient for many subjective tasks such as toxicity and hate-speech detection. Polarization has been proposed as a robust mechanism that discerns minor disagreements from systematic, fundamental differences of opinion, but there exist no practical tools that attribute polarization to annotator subgroups. We evaluate current approaches, discovering severe flaws in practical settings; (1) the existence of “inherent” polarization that is unattributable to any known or latent groups, and (2) opposing directions in polarization canceling out aggregated annotations. Thus, we introduce a new metric which quantifies and checks the statistical significance of attribution of polarization to any group, while circumventing these limitations. We apply our metric to four subjective NLP datasets, finding generalizable results across all datasets, and validating results from prior literature. Finally, we estimate the minimum number of annotators needed to obtain robust results, and provide an open-source Python library that implements our metric.

1 Introduction

Annotations are essential for a wide range of tasks, especially in fields with subjective judgements, such as content moderation, toxicity, and hate speech detection. These tasks are especially important for training supervised learning models that remove or flag comments targeting vulnerable and minority groups in online spaces, where they are often vulnerable (United Nations, 2025; Dioum, 2025). However, annotations for such tasks are subjective and typically aggregated based on majority-voting (Ivey et al., 2025; Giannoutsos et al., 2025; Kralj Novak et al., 2022; van der Velden et al., 2025). This may at best limit the usefulness of the data, or at worst lead to biased and misconfigured systems (Sap et al., 2019; van der Velden et al., 2025).

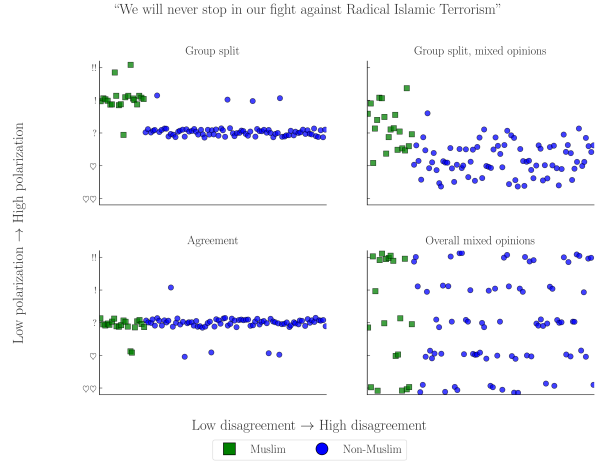


Figure 1: Simulated experiments for four scenarios on a hate speech detection task. Depending on the annotators, there may be a clear pattern where the split in annotations (polarization) can be attributed to a specific group (Muslim annotators in the case of the upper plots).

The core of the problem is simple; if a comment targets marginalized groups, there can be cases where most annotators overlook it, while the few marginalized annotators vehemently disagree with the majority. These *fundamental* disagreements are typically ignored via aggregation (e.g., majority label voting) or filtering (e.g., keeping only items with large overall annotator agreement). For example, coded language may not be picked up by most annotators but only by ones belonging in the targeted minority (Quaranto, 2022), as shown in Fig. 1. Furthermore, majority-group annotators may themselves be biased against minority groups, e.g., toxicity models disproportionately marking African American speech as “toxic” (Sap et al., 2019).

Instead of using inter-annotator agreement, we could use *polarization*, which formulates annotation analysis as a cluster identification problem (Pavlopoulos and Likas, 2023, 2024). While we can detect polarization (Pavlopoulos and Likas, 2024, 2023), there is little work on how to detect whether it is *actually caused by marginalized groups*. Our work thus tries to answer the following Research Question:

RQ. *Can we attribute polarization to annotator subgroups?*

We first identify limitations of current work on polarization:

1. Existing mechanisms (Pavlopoulos and Likas, 2024) operate only as binary classifiers, producing non-quantifiable/comparable results.
2. These mechanisms only test for the extreme case where *all* present polarization can be attributed to a single annotator group, which we find to be extremely rare in real-life datasets (App. B).
3. We are the first to report the existence of *inherent polarization*: most comments having some degree of polarization that cannot be attributed to *any possible partition* of annotators (Finding 1).
4. We also find that annotator groups may exhibit opposing polarization attribution between comments, requiring isolating annotations to each individual comment when analyzing whole datasets (Finding 2).

We introduce a new metric (“Aposteriori Unimodality–*apunim*”) that attributes polarization to specific annotator groups without and develop a parametric test for statistical significance. Our metric is generalizable across any domain and modality (e.g., video/image classification) and, unlike previous approaches, enables *quantitative* comparisons across datasets. By considering the entire set of comments in a dataset *apunim* helps to identify genuine polarization and distinguish it from chance attribution, which is essential when analyzing large datasets.

Having established our metric, we apply it to four Natural Language Processing (NLP) datasets focused on Toxicity and Hate Speech detection (§5), verifying current literature on the different annotation patterns of sociodemographic and ideological subgroups (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022). Finally, we release an open source Python library (Appendix D) that implements our metric, which can be directly installed via PyPi.¹ and provide an estimation on the minimum number of annotators required for robust estimates (§6). The code for the experiments, graph creation, and analysis can be found on our project’s repository.²

¹Library: <https://pypi.org/project/apunim>
Documentation: <https://dimts-ts.github.io/apunim>

²Replication code: <https://github.com/dimts-ts/aposteriori-unimodality>

2 Related Work

2.1 Agreement

There are many popular metrics for measuring inter-annotator agreement such as Cohen’s kappa, Krippendorff’s alpha and Fleiss’s kappa. All of these metrics attempt to distinguish genuine disagreement among annotators from noise (or “random disagreement”) using statistical methods. However, most cannot be generally used for comparing intergroup annotation agreement without extending them (e.g., computing pairwise Cohen’s kappa for all groups), or placing constraints on when they can be used (e.g., no missing labels and same number of annotations per annotator for Fleiss’s kappa). They have also been criticized for ignoring fundamental differences that can arise from different social, cultural and demographic backgrounds (Feinstein and Cicchetti, 1990; Byrt et al., 1993; van der Velden et al., 2025; Checco et al., 2017), for their inability to quantify and compare results across experiments (Wong et al., 2021), and for fundamentally faulty assumptions around the concept of “chance-correction” (Checco et al., 2017).

Recent work has attempted to work around these limitations, by working with simpler statistical models (van der Velden et al., 2025), adapting Cohen’s kappa for subgroup agreement (Wong et al., 2021), or creating a Bayesian model to explicitly differentiate between subgroup disagreement and noise (Ivey et al., 2025). While these extensions do improve the performance of these metrics, they are typically less explainable and intuitive.

2.2 Polarization

There are multiple competing formulations for polarization. One of these approaches (Akhtar et al., 2019) utilizes χ^2 to estimate in-group vs out-group variance, but still necessitates traditional agreement metrics for analysis. Others attempt to solve this problem by framing disagreement as a cluster identification problem, applying either parametric (Checco et al., 2017), or non-parametric (Pavlopoulos and Likas, 2024; Mignemi et al., 2024) approaches to the annotation histograms.

For our work, we use a non-parametric approach, normalized Distance From Unimodality (nDFU) (Pavlopoulos and Likas, 2024), which measures the degree of polarization in a distribution of opinions. The metric takes values in the range $[0, 1]$, where values close to 0 indicate little or no polarization (i.e., opinions are concentrated around a single mode), while values close to 1 indicate strong polarization with multiple distinct peaks. The metric is defined as follows:

$$\text{nDFU}(X) = \frac{\max_{i \neq m} \left(f_i - \begin{cases} f_{i-1}, & \text{if } i > m \\ f_{i+1}, & \text{if } i < m \end{cases} \right)}{f_m} \quad (1)$$

where $m = \arg \max_i f_i$

Here, X denotes the set of annotations, and f_i the relative frequency of the i -th annotation, and m the histogram peak. Intuitively, nDFU measures how prominent secondary peaks are compared to the main peak.

2.3 Attributing polarization to groups

While many studies study the effect of subgroups on annotation agreement (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), attributing polarization to subgroups is a relatively new idea. Under the name *aposteriori unimodality* (Pavlopoulos and Likas, 2024), the original idea is that if polarization in a single item is positive, but zero for all annotator subgroups, then those subgroups are “causing” the exhibited polarization. Though promising, this mechanism operates on an edge-case scenario where *all* polarization is explained by a single split. Additionally, getting polarization estimates from few annotations per item is extremely noisy, which is especially problematic when the output is binary (i.e., an item either is or is not polarizing) instead of a quantifiable value.³

3 What are the limitations of polarization attribution?

3.1 Problem Formulation

Let $d = \{c_1, c_2, \dots\}$ be a dataset composed of multiple annotated comments.⁴ Each comment c is assigned multiple annotators, which creates the annotation multi-set $A(c) = \{(a_1, g_1), (a_2, g_2), \dots\}$, where a is a single annotation. As an example, the annotations for a comment in a sentiment analysis task where we group the annotators by gender, would be formulated as:

$$A(\text{“Could be better, could be worse”}) = \{ (+, F), (-, M), (+, F), \dots \}$$

Intuitively, a group partially explains the polarization of a comment c when the annotations of that group $A(c | g) = \{a(c; g') \in A(c) | g' = g\}$ exhibit higher polarization compared to the full

³For example, when presented with noisy estimates of a metric in the $[0, 1]$ range, we would ideally want to distinguish between cases where attribution is close to 0.001, and cases where attribution is 0.49.

⁴These items could be of any modality, such as comments in a discussion, images, videos or survey questions, even though we consider comments in this study.

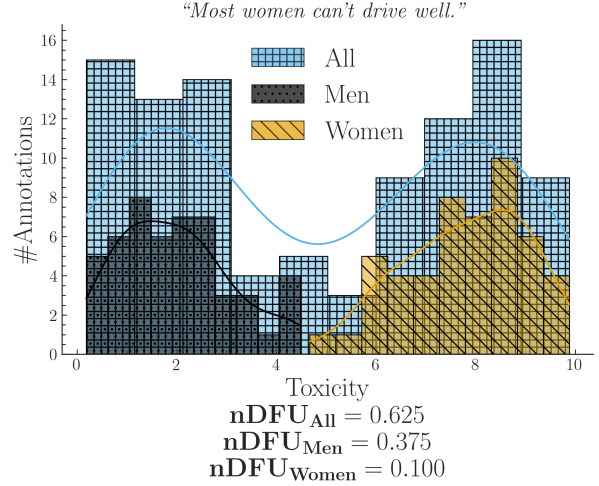


Figure 2: Synthetic experiment simulating a polarizing comment, where male and female annotators agree between themselves, but disagree with the opposite gender.

set of annotations $A(c)$. Consider the (simulated) case where a misogynistic comment is annotated for toxicity by male and female annotators shown in Fig. 2. The annotations are generally polarized ($\text{nDFU}_{all} = 0.62$); but both female ($\text{nDFU}_{women} = 0.10$) and male ($\text{nDFU}_{men} = 0.37$) annotators exhibit low polarization between their respective groups. This pattern suggests that the overall polarization is driven by substantive disagreements between male and female annotators.

3.2 Inherent polarization

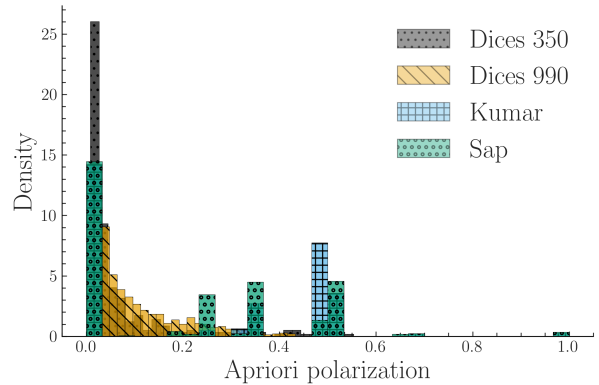


Figure 3: Apriori (inherent rather than observed) polarization across our datasets. Contrary to assumptions made in prior work, inherent polarization is rarely zero and can even reach $\text{nDFU} = 1$.

Inherent polarization refers to disagreement that cannot be attributed to *any specific* grouping of annotators; that is, polarization that exists no matter how we split the annotators in any arbitrary group. Prior work (Pavlopoulos and Likas, 2024)

attributed polarization to annotator groups only when the observed polarization for each individual group became exactly zero (e.g. $nDFU_{all} > nDFU_{men} = nDFU_{women} = 0$). The presence of inherent polarization would invalidate this theory, since there would be no possible group g s.t. $nDFU_g = 0$.

To examine this assumption, we create random annotator groupings for each comment in our datasets. Specifically, for every comment, we randomly partition the annotators into random groups (with randomly selected sizes) 100 times and compute the observed polarization for each partition. We then take the *minimum* observed polarization scores across all random partitions, which guarantees that we check for all possible variables, known and latent.

Formally, the inherent polarization for a single comment c is given by:

$$Pol_{inh}(c) = \min_i \{nDFU(\tilde{r}_i(A(c)))\} \quad (2)$$

where $\tilde{r}$ is the random partition operator.

Repeating this process for every comment produces the distribution shown in Fig. 3, demonstrating that inherent polarization is consistently non-zero and can even be substantial for many comments.

Finding 1. *Each comment may exhibit a degree of inherent polarization that cannot be attributed to any known or latent annotator grouping.*

3.3 Conflicting polarization

Previous work (Pavlopoulos and Likas, 2024) stopped on a comment-to-comment analysis. In practice however, this approach is inherently noisy due to a lack of information (i.e., limited annotations) relating to each individual comment. Many datasets in NLP are annotated by only 3–5 annotators per comment, which is especially problematic when we want to analyze patterns in annotator subgroups; in a best-case scenario of 5 annotators split evenly between two groups, we would get a 3-2 split. This issue is especially prevalent for polarization, since $nDFU$ is *undefined for groups of two* (and arguably marginally useful for groups of three), since it requires at least two histogram peaks to function (see Eq. 1), a limitation not acknowledged in the original work.

This limitation leaves us with two choices: (1) only use datasets featuring at least 6–8 annotators, and (2) aggregate annotations across comments for each dataset. The former case is unrealistic, especially for most NLP datasets due to annotation costs. We consider the latter case in an experiment shown in Fig. 4 where we combine annotations across two comments. We observe that, should the two comments be polarized for the opposite reason, the opposing polarization effects might bal-

ance each other out, leading to a false negative. Therefore, we should never aggregate annotations across comments.⁵

Finding 2. *Polarization has a direction, which may compromise cross-comment annotation aggregation.*

4 How do we fix it?

4.1 From inherent to apriori polarization

In §3.1, we explained how our methodology needs to compare the polarization of $A(c)$ and $A(c | g)$. Direct comparisons are not advisable, since we would be comparing statistics between a set and its subset—in this case, standard practice dictates comparing the subset with its complement; i.e., the in-group vs. the out-group instead of the in-group vs. all the annotations. However, this also would not work; if we apply the subset-vs-complement method in the example of Fig. 2, we would end up subtracting the $nDFU$ s of men and women, which are similar as they are both unimodal distributions, concluding that there is no polarization present.

We instead compare the observed polarization of $A(c)$ with its inherent polarization, following the methodology introduced in §3.2 with two key distinctions. (1) Instead of using random splits, we create random stratified splits.⁶ (2) Instead of finding the minimum possible polarization exhibited by any group, we calculate the average. Therefore, we do not test whether the group is polarized against the whole; nor whether the group explains *some polarization* (as would be the case if we used Eq. 2), but rather whether the group explains polarization *more than other groups*, which is a much stricter question.⁷ Specifically, we define a-priori unimodality for a comment c as:

$$Pol_{apr}(c) = \frac{1}{t} \sum_{i=1}^t nDFU(\tilde{r}_i(A(c))) \quad (3)$$

4.2 Expanding to the whole dataset

As discussed in §3.3, we must not aggregate the annotations themselves before applying the $nDFU$ metric. We can however estimate how much polarization is generally present in the dataset, by simply averaging the *a-priori* polarization of its comments. Intuitively, a high apriori polarization score means that much of the polarization in the

⁵A similar assumption was followed in early parametric approaches, which assumed different parameters for each comment (Checco et al., 2017).

⁶I.e., if we have 100 annotations, 80 of which are made by male annotators and 20 by female annotators, we create random partitions of sizes 80 and 20.

⁷Indeed, by definition, all groups exhibit equal or larger polarization than the inherent pol. value.

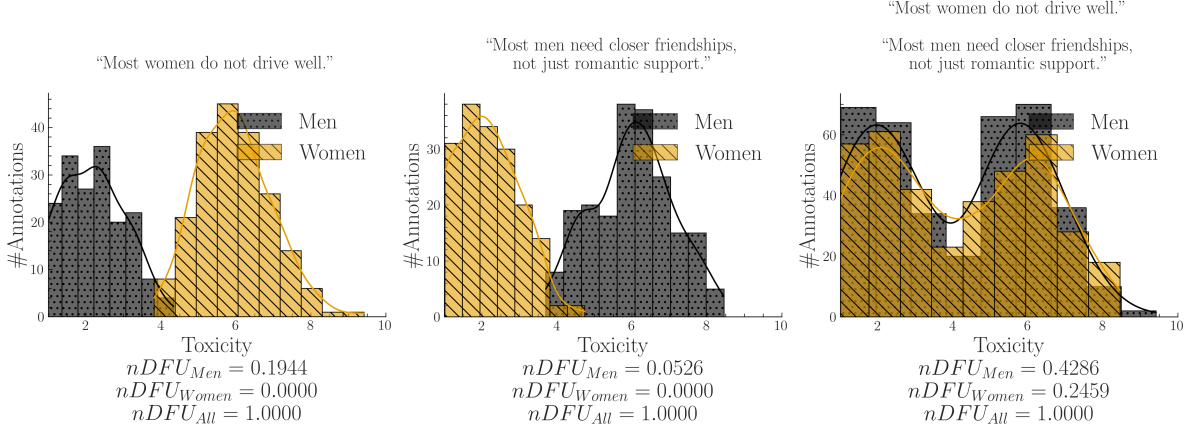


Figure 4: A synthetic experiment simulating a polarizing discussion with two comments where annotators disagree about which one is toxic. When we aggregate the comments, we unintentionally change the comparison: instead of comparing two unimodal distributions (each comment viewed separately) with two bimodal distributions (their aggregated annotations), we end up comparing two bimodal distributions, obscuring the source of the polarization.

dataset is likely not driven by any group.

$$Pol_{apr}^d = \frac{1}{|d|} \sum_{c \in d} Pol_{apr}(c) \quad (4)$$

Which we will compare with the average observed polarization of all comments:

$$Pol_{obs}^d(g) = \frac{1}{|d|} \sum_{c \in d} nDFU(A(c | g)) \quad (5)$$

We can now compare the polarization that is attributed to a single-dimensional annotator group (Equation 5) to a prior one (Equation 4). We define our normalized metric, which we call “*apunim*” (*A posteriori Unimodality*), as:

$$apunim(d, g) = \frac{Pol_{obs}^d(g) - Pol_{apr}^d}{1 - Pol_{apr}^d} \quad (6)$$

Since $nDFU \rightarrow [0, 1] \Rightarrow apunim \rightarrow [-1, 1]$. The interpretation of this metric boils down to three scenarios:

- $apunim(\theta) \approx 0$: The examined group does not meaningfully disagree with the other annotators.
- $apunim(\theta) > 0$: The examined group has fundamental disagreements with the other annotators.
- $apunim(\theta) < 0$: Overall polarization *drops* when we add annotators from this group to this task. This occurs when intragroup polarization is higher than apriori polarization; indicating that annotators of that group are polarized between themselves.

4.3 Statistical significance

Since the underlying annotation histograms may be noisy, we develop a statistical significance test.

Dataset	#Com	#Ann	#Dims	Task
Kumar	106,035	5	10	Tox
Sap	626	5–6	3	HS
DICES-350	72,103	70–76	4	HS
DICES-990	43,050	123	4	HS

Table 1: Datasets used in this study. $\#Ann$ refers to the number of annotations per comment (see App. B), and $\#Dims$ to Personal Characteristic (PC) dimensions. Tasks are either toxicity (Tox) or hate speech detection (HS).

If the *apunim* value is statistically significant, we expect it to significantly differ from *apunim* values exhibited by random annotator groups. We use the individual randomized partitions $r_i(A(c))$ to compute pseudo-*apunim* values, which are then compared to the actual *apunim* value of the sample using a Student-T test. Thus, we test a much stricter hypothesis:

$$H_0 : apunim(g) = apunim(d)$$

$$H_a : apunim(d, g) \neq apunim(d)$$

where $apunim(d)$ denotes the mean *apunim* values of each random partition $\tilde{r}_i(A(c))$. For a detailed explanation of the p-value implementation, as well as a discussion around the assumptions made about the test, refer to App. C.

5 Attributing real-life polarization to human groups

5.1 Datasets

We use four datasets from current NLP research that include PC information at the level of individual annotators; specifically, the “Kumar”(Kumar

et al., 2021) and “Sap”(Sap et al., 2022) datasets, which are concerned with Toxicity and hate speech detection respectively, and the DICES-350 and DICES-990 datasets (Aroyo et al., 2023), which focus on LLM response safety.⁸ While the toxicity and hate speech detection tasks are not equivalent, they share substantial conceptual and operational overlap, as both require annotators to assess the presence and severity of harmful or abusive language. Thus, we expect the sources of annotator disagreement and bias that arise in these tasks will be broadly similar.

The datasets differ substantially in scale and annotation design. The Kumar dataset stands out for its large size, featuring more than 100,000 annotated comments, as well as 10 distinct PC dimensions. In contrast, the DICES datasets contain fewer annotated comments but involve more than an order of magnitude more annotators per comment than the other datasets. A brief overview of dataset characteristics is provided in Table 1. The Kumar and Sap datasets contain a large number of polarizing items, unlike the DICES datasets, where most items are unpolarized (Fig. 7).⁹

5.2 Experimental Setup

Assuming a confidence level of $\alpha = 0.05$ we test whether any of the provided PC dimensions can partially explain polarization in the toxicity / hate speech annotations, as shown in Table 1.¹⁰ Note that some groups are not included, either because the results were not statistically significant, or because there were no polarized items annotated by more than one annotator of these groups (see §??). See App. E for details on other parameters and execution times, and App. F for the full results.

5.3 Which factors contribute to polarization?

Across all four datasets, we observe a consistent pattern in which annotation polarization is linked to annotator characteristics, particularly ethnicity. In the hate speech detection datasets (DICES-350, DICES-990, Sap), *ethnicity is the strongest contributor to polarization*, with statistically significant effects for at least one major ethnic group in each dataset (African Americans (AAs): -0.0428 /Asians: 0.0659 for DICES-350, AAs: -0.2967 /Whites: 0.1400 for Sap, Whites: -0.0377 for Kumar, and all ethnicities

⁸For the latter, we consider only labels relating to racism (Q3_bias_overall), thereby transforming the task to hate speech detection.

⁹This systematic discrepancy could be attributed to the DICES datasets being composed of Large Language Model (LLM)-generated dialogues with already-aligned models tuned to avoid racist speech.

¹⁰We sample 10,000 comments from the Kumar dataset due to computational costs.

Group	apunim	support
DICES-350		
Race=AfricanAmerican	-0.043	9077
Race=Asian	0.066	8138
DICES-990		
Age=GenX+	-0.047	14384
Age=GenZ	0.064	13741
Edu=High school and below	-0.075	7419
Gender=Man	0.028	32494
Gender=Woman	-0.024	33471
Race=AfricanAmerican	0.091	5810
Race=Latino	-0.133	6610
Race=Other	0.075	24321
Race=White	-0.120	11392
Sap		
Eth=Black	-0.297	1041
Eth=White	0.140	2048
Gender=Male	0.094	1635
Gender=Female	-0.181	1333
Kumar		
Edu=Masters	0.015	13420
Edu=Bachelors	0.040	34085
Edu=Associate degree	-0.011	9122
Edu=None	-0.034	16666
Eth=White	-0.038	45306
Gender=F	-0.028	40686
Gender=M	0.028	37929
PastTargeted=No	-0.063	45234
PastTargeted=Yes	0.035	24491
Parent=No	-0.071	37231
Parent=Yes	0.054	41493
Trans=No	-0.089	11978
Politics=Conservative	0.033	23010
Politics=Independent	-0.016	22818
Politics=Liberal	-0.029	32917
ReligionImportant=No	-0.080	26008
ReligionImportant=Low	-0.015	10055
ReligionImportant=Mid	0.021	20169
ReligionImportant=High	0.049	27284
SeenToxicity=No	0.031	20185
SeenToxicity=Yes	-0.072	43265
SexOrientation=Bi	0.018	10005
SexOrientation=Het	-0.057	38048
TechImporant=Mid	-0.027	40477
TechImporant=High	0.021	22471
ToxProblem=Rare	0.018	14457
ToxProblem=Freq	-0.022	24774
ToxProblem=VeryFreq	-0.015	10656

Table 2: Statistically significant ($p < 0.05$) a-posteriori unimodality results across all four datasets. High *apunim* values indicate that polarization can be attributed to this group, while negative values (gray rows) indicate that there is more polarization among annotators of that group compared to all annotators (see §4.2). Consult App. F for the full results.

for DICES-990). These results align with prior work suggesting that ethnic background is a systematic source of annotator disagreement rather than isolated noise (Goyal et al., 2022), validat-

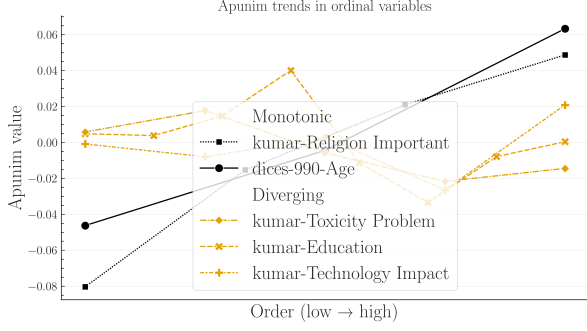


Figure 5: Apunim values across ordinal factor levels with at least two statistically significant values. The x-axis corresponds to the ordered levels of each factor and is normalized so that different ordinal scales are comparable (e.g., education which is measured in different scales across datasets). The left side of the x-axis (low orders) refers to young annotators, low education, low religiousness, and indifference towards the impacts of toxicity and technology. The full ordinal modes for each dimension and their individual labels can be found in F.

ing our initial motivation. Similarly, less educated annotators tend to show higher subjectivity, with negative polarization contributions in DICES-990 (-0.0751) and lower contributions from college-educated groups (Table 2). Overall, our results support prior studies on how annotator characteristics—including ethnicity, education, gender, and personal experiences—shape annotation patterns (Goyal et al., 2022; Binns et al., 2017; Giorgi et al., 2025; Al Kuwatly et al., 2020; Sachdeva et al., 2022), confirming that polarization is influenced by systematic factors rather than random noise.

5.4 Why we should stop binarizing groups

Analysis on annotated data frequently ignores the different kinds of measurements in studies using sociodemographic and LIKERT-style attributes. One could treat all dimensions as *nominal/categorical*, even if they are inherently ordinal (e.g., LIKERT-scale, education level) or even binarize when many values exist (e.g., age). With this approach, however, not only do we lose valuable information, but we may magnify statistical errors (Streiner, 2002). By contrast, we posit that *retaining the order of ordinal attributes reveals interesting trends* that would be otherwise missed with categorical inference.

Figure 5 shows *apunim* trends per ordinal PC attribute as shown in Table 2, focusing only on attributes with at least two statistically significant values ($p < 0.05$). This condition is necessary, since we cannot infer trends from dimensions with

a single non-zero *apunim* value.¹¹ We can see that **Religion Importance** is characterized by a strict monotonic increasing trend. This indicates that non-religious annotators (low order) are more divided among themselves, while very religious annotators (high order) agree among themselves, but fundamentally disagree with other annotators. A very similar, inverse trend is observed for **Age**. For **Education**, we observe a peak in the middle (4—“Bachelor’s Degree”), which indicates two findings. First, up to an education level and as their level increases, people tend to agree more with people from a similar level. Second, beyond that level and till they reach the highest level, people tend to disagree more on their own level compared to ones from a different level.

6 How many annotators do we need?

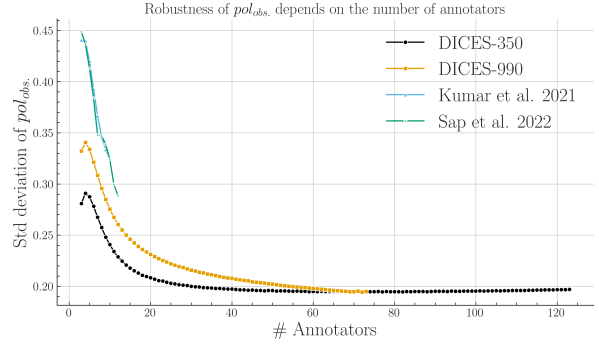


Figure 6: Standard deviation of 30 randomly sampled (with replacement) pol_{obs} values (Eq. 5). The x-axis shows the sample size (number of annotators), starting from 3 and increasing one at a time, up to the largest annotator count for which a sufficient number of items exists (B).

While our test works best with many annotations per item, the cost required for multiple human annotators is often greatly restrictive (Rossi et al., 2024). Thus, it is imperative to find the least annotators needed for *apunim* to generate stable and reliable estimates. For each item per dataset, we randomly sample with replacement a subset of annotators and compute the standard deviation of 30 sampled pol_{obs} values (see Equation 5). Figure 6 demonstrates the effect of the number of annotators on the reliability of the polarization estimation, which we discuss next.

Each dataset features a different standard deviation even with the same number of annotators,

¹¹Note that **Age** is typically a ratio (positive, continuous) variable, although it is treated either as ratio (Sap), interval (Kumar), or ordinal (DICES) variable depending on the dataset. Since ratio and interval variables are also ordinals, we treat **Age** as ordinal in this analysis.

which is broadly consistent with the amount of polarization in each dataset (Appendix B). However, we observe a clear, non-linear, inverse relationship between the number of annotators and the standard error across datasets. This can be attributed to a more robust histogram estimation used by nDFU (§??) when provided with more points for each of the subgroups. In the case of our datasets, we observe that after about 20 points, the estimation is good enough that more annotations contribute trivially to the robustness of the metric.

7 Conclusion & Future work

We introduced the *apunim* metric which attributes polarization to annotator subgroups for various PC dimensions. Our metric works well for imbalanced annotator splits, is well-defined for multi-label annotation, includes a statistical significance test, and unlike previous work, quantifies the amount of attribution in order to allow for comparisons between various experiments and datasets. Using this metric, we discover that annotator ethnicity is the strongest and most consistent driver of polarization between annotators across datasets, and is much more pronounced on hate speech detection tasks. We also discover that education and religion consistently impact the behavior of annotators. Specifically, we discover that very religious annotators fundamentally agree with themselves but disagree with others in Toxicity detection tasks, a trend that is gradually and consistently diminished, and ultimately reversed, by irreligious annotators. We also provide an open-source, PyPi-installable implementation, and an estimation of the number of annotators needed for each item, for our method to be consistently reliable.

Our framework has broad applications in NLP sub-tasks relating to content moderation (such as toxicity and hate speech detection). Since many such tasks are run on multiple discussions, our framework can be used on the discussion, instead of the dataset level. In that case, researchers may be able to isolate polarizing discussions, and investigate underlying patterns of polarization for individual minority groups. Alternatively, items can be grouped by comment turn, allowing researchers to investigate the evolution of polarization across discussions. Lastly, our work can be used to more reliably test whether LLM-as-a-judge systems can replicate the polarization patterns of certain minority groups.

8 Limitations

Researchers often only publish single-label, aggregated data, or datasets without annotator PC characteristics. Our analysis and the applicability of our framework is thus constrained by the lack of

availability of such datasets. Additionally, while the findings presented in §5 do generally agree with current literature, the polarization patterns observed should be verified with more datasets before any generalizable claims are made.

9 Ethical Considerations

All datasets used in this study were either publicly accessible, or were provided to us by the authors with full disclosure on the purpose of our research. No identifiable, personal information is included in our pipeline. We did not perform any annotation tasks ourselves.

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Appendix A Acronyms

LLM	Large Language Model
PC	Personal Characteristic
nDFU	normalized Distance From Unimodality
FWER	Family-Wise Error Rate
NLP	Natural Language Processing
GPLv3	GNU General Public Licence v3
AA	African American

Appendix B Annotation details

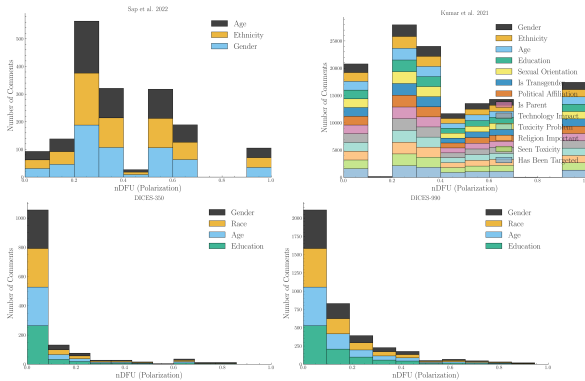


Figure 7: Histogram of dataset polarization for the human datasets considered in this paper per PC.

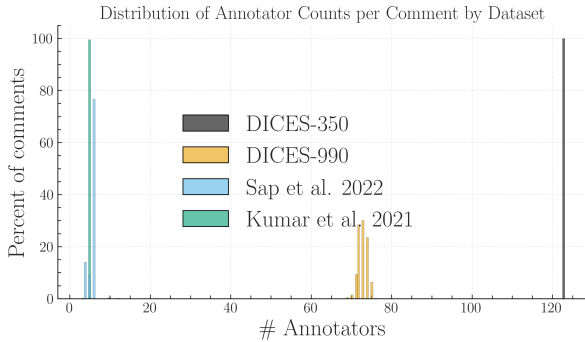


Figure 8: We calculate the ratio of comments for each dataset (y-axis) that had x number of annotators (x-axis). We truncate the x-axis to 130 annotators, disregarding very few comments in the Kumar dataset exceeding this threshold (see Table 3).

The number of annotators per comment for all datasets presented in §?? can be found in Figure 8, and descriptive statistics can be found in Table 3. The DICES-990 dataset has exactly 123 annotations per comment, while the other datasets exhibit a few variations around the central mean reported in Table 1. Interestingly, the sample used from the Kumar dataset exhibits a few comments

Table 3: Descriptive statistics for the number of annotations per comment, grouped by dataset.

	count	mean	std	min	25%	50%
DICES-350	350	123	0	123	123	123
DICES-990	990	72.8313	1.1652	69	72	73
Kumar et al. 2021	20000	5.0925	4.8872	5	5	5
Sap et al. 2022	585	5.6359	0.7713	3	6	6

that vastly exceed the promised 5 number of annotators. We included these comments for the experiment presented in Table 2, but not for the annotator number experiment in Figure 6 since they would produce uninterpretable figures.

Appendix C P-value estimation implementation

Since our test is parametric, it assumes that there exist (1) enough data-points to reliably run the means-test, (2) enough annotations per PC group to estimate nDFUs in Equations 3-5, (3) the apunims of the random partitions are independent and identically distributed. Condition (1) can be safely assumed for most datasets, because they feature enough items to reliably assume normal residuals (see §??). We discuss condition (2) in §6. Condition (3) is partly fulfilled; the partitions are identically distributed, but may be dependent on the sample distribution.

Replacing the parametric means test with a non-parametric alternative would relax assumptions about the underlying distribution but would not resolve the potential dependence between the random apunims. We initially attempted to implement a permutation test on the empirical (random apunim) distribution, but this resulted in excessive computational costs.

While the test operates on a single θ dimension in principle, in practice we often want to test polarization for all subgroups ($\forall \theta \in \Theta$). Since we are simultaneously considering $|\Theta|$ hypotheses, we apply a multiple comparison correction to the resulting p-values—specifically the Holms method (Holm, 1979). The test is parameterized by the Family-Wise Error Rate (FWER), which is used to tune the strength of the correction; we can increase this value to make our test more conservative towards multiple hypotheses (Chen et al., 2017)). In general, it is safe to set FWER equal to the significance level of our test (e.g., $FWER = 0.95$ if we are looking for $p < 0.05$).

Appendix D Software

D.1 Installation and usage

We release the `apunim` package, which is a Python library that implements both the nDFU and

apunim calculations, as well as its statistical test. The software is available on PyPi (<https://pypi.org/project/apunim>). We also provide high-level documentation in HTML format that includes an introduction, guide, and full low-level documentation for each of the two provided functions (<https://dimits-ts.github.io/apunim>).

The code is open-source, licensed under the GNU General Public Licence v3 (GPLv3) and can be found on the project’s repository (<https://github.com/dimits-ts/apunim>). The library is cross-platform, works for any Python 3 version, and requires very few dependencies (namely `numpy`, `scipy`, `statsmodels`).

D.2 Implementation details

By default, the computation of the `apunim` values happens on a single thread. In cases of serious computational cost, we recommend running the `apunim` calculation on multiple concurrent processes, since all operations are CPU-bound and thread-safe. This may be useful in cases where there are multiple `PC` dimensions, each of which is computationally expensive.

In order to lower the computational cost required, we re-use the random partitions $\tilde{P}_i$, which are originally calculated to obtain the apriori polarization values (see Equation 3), for the p-value calculations (§??). This means that in our current implementation, the number of groups for estimating apriori polarization ($\tilde{P}_i$ in Equation 3), and the number of `apunim` values used in the p-value calculation ($\tilde{D}_i$ in Algorithm ??) have to be the same.

Appendix E Replication Details

We use version 1.0.1 of the `apunim` library for our experiments. We use 100 random iterations for the apriori value calculation (Equation 3) and the p-value estimation (§??), and set a `FWER` rate of 0.95.

The current version of the library handles large number of annotators very well due to the use of vectorized operations, but struggles when faced with a lot of comments when the annotators belong to multiple groups. Indeed, while the experiments for all datasets presented in §?? need no more than a minute to run, the Kumar dataset requires about 12 hours of computation even with a 20% sample. The annotator variance analysis presented in §6 also required more than 10 hours of computation, although that is expected due to the nature of the experiment, which is very computationally demanding, and a non-standard use of our library.

This bottleneck is likely caused by repeated creation of `numpy` arrays, resulting in repeated memory allocations for each subgroup, in each comment, for each `PC` dimension. We plan on pro-

filing and addressing it in the next releases of our software. Until then, we suggest either using multiprocessing for each `PC` subgroup, or running the different experiments in parallel (indeed, we use the `gnu-parallel` linux package (Tange, 2025) for our own experiments).

In Figure 6, we removed values where there was a lack of items annotated by the specified number of annotators (e.g., only 7% of comments in DICES-990 were annotated by more than 73 annotators—see B).

Appendix F Full Results

Table 4: Aposteriori unimodality results for the DICES-350 dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	0.0186	9703
	2) Millenial	-0.0059	11268
	3) Gen. Z	-0.0066	17528
Education	College or higher	0.0124	23475
	High school or lower	-0.0152	12833
	Other	-0.0214*	2191
Gender	Man	-0.0193	19093
	Woman	0.0215	19406
Race	African Am.	-0.0428	9077
	Asian	0.0659	8138
	Latino	-0.0030	6886
	Multiracial	-0.0001	5008
	White	-0.0198	9390

Table 5: Aposteriori unimodality results for the DICES-990 dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	-0.0471	14384
	2) Millenial	-0.0043	38436
	3) Gen. Z	0.0644	13741
Education	College or higher	0.0095	59142
	High school or lower	-0.0751	7419
Gender	Man	0.0278	32494
	Woman	-0.0235	33471
Race	African Am.	0.0909	5810
	Latino	-0.0030	6886
	Multiracial	-0.0001	5008
	White	-0.0198	9390

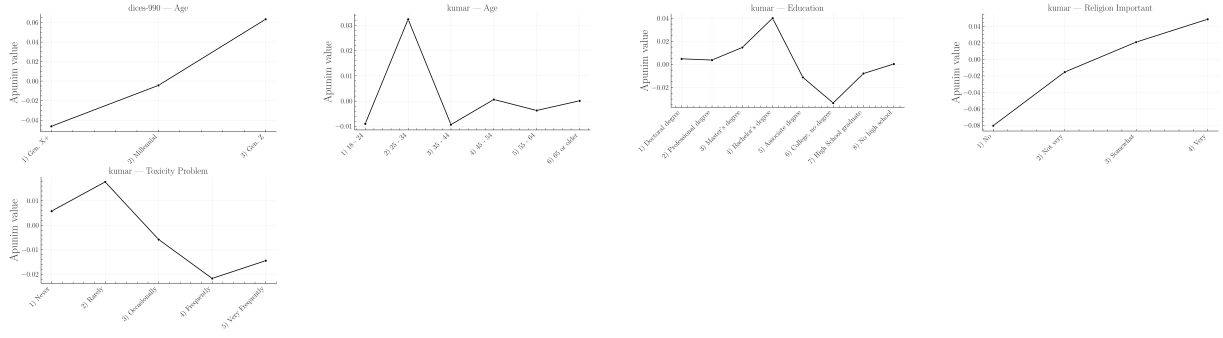


Figure 9: Apunim values across ordinal factor levels with at least two statistically significant values, expanded for each ordinal PC. This figure is complementary to Figure 5. The values in the graphs were extracted from the Tables of F.

Table 6: Aposteriori unimodality results for the Sap dataset.

PC dimension	Value	apunim	support
Age	1) Gen. X+	0.0192	743
	2) Millennial	0.0189	1792
	3) Gen. Z	-0.0153	239
Ethnicity	Black	-0.2967	1041
	White	0.1400	2048
Gender	Male	0.0942	1635
	Female	-0.1810	1333